

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a7316c5ee2dcba4ec.jsonl`  
- SHA-256: `ee549cfa31c4299b8568f595c498de72d10993f9f8eac3f4860fbf7b74327584`  
- Source modified: `2026-06-06T17:37:47+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:36:30.232Z)

Adversarial Claim Verifier (voter 2/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"A two-stage audio pipeline (energy-based peak detection followed by a small CNN) can detect racket-ball contact events from sound with near-millisecond temporal precision (energy peaks detected on average 0.2 ms after onset, even with noise), demonstrating that audio is a high-resolution local cue for hit timing in a racket sport."

****Source:**** <https://arxiv.org/html/2409.11760v1> (primary)

****Supporting quote:**** "The detection algorithm shows high temporal resolution, detecting energy peaks on average no later than 0.2 ms after occurring... we use energy-based peak detection to identify potential bounce sounds. Then, a CNN classifies the detected peaks."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:36:35.562Z)

I'll verify this claim by examining the source carefully. Let me fetch the primary source first.

3. assistant (2026-06-06T17:36:37.891Z)

Let me fetch the primary source to examine the claim in context.

4. user (2026-06-06T17:36:48.922Z)

Table Tennis Audio Detection Research Summary

****Sport:**** Table tennis

****Audio Detection Task:**** Detecting ball bounce sounds and classifying them by spin type and surface/racket characteristics.

Specific Answers:

**** (1) Temporal Resolution (0.2 ms claim): ****

The 0.2 ms refers to detection latency. The paper states: "The detection algorithm shows high temporal resolution, detecting energy peaks on average no later than 0.2 ms after occurring."

This measures when energy peaks are detected relative to their actual occurrence. The task involves detecting the ****racket-ball contact sound**** specifically?the paper explains that "When the ball bounces against a table tennis racket, the contact duration ranges from 1.3 ms to 1.8 ms."

(2) Detection Method:

Yes, exactly correct. The approach uses "energy-based peak detection to identify potential bounce sounds. Then, a CNN classifies the detected peaks."

(3) CNN Classifier Accuracy:

The CNN achieved:

- **Spin classification:** 95% recall, 98% precision, 96% F1-score
- **Surface classification:** 97% recall, 97% precision, 97% F1-score

(4) Audio Source:

The audio captures the **racket-ball contact sound**, not subsequent bouncing. The paper focuses on "the sound generated when the ball strikes the racket" as the information source for spin detection.

5. assistant (2026-06-06T17:36:51.969Z)

The source confirms several aspects of the claim. Let me dig deeper into the specifics, particularly whether "0.2 ms after onset, even with noise" is accurate and what the energy peak detection actually measures, since this is the load-bearing part of the claim.

6. user (2026-06-06T17:37:00.636Z)

Energy-Based Peak Detection Details

1. Temporal Resolution Measurement

Exact quote on the 0.2 ms result:

"The detection algorithm shows high temporal resolution, detecting energy peaks on average no later than 0.2 ms after occurring."

How it was evaluated: The paper tested this on two conditions?clean audio and noisy audio with human speech overlaid. The table shows results for "No Noise" (0.09 ms onset accuracy) and "Noise" (0.2 ms onset accuracy), indicating the noisier condition produced the slower detection.

2. Noise Robustness Claims

Direct quote on robustness:

"These results remain consistent even with human speech, recorded in the same condition, being laid over the data, simulating a noisy environment."

The paper explicitly claims the method maintains performance with background speech noise, though detection latency increases from 0.09 ms to 0.2 ms under noise.

3. Sampling Rate

The audio was recorded at **44.1 kHz**, as stated: "recorded with the Zoom H4n Pro Handy Recorder...at 44.1kHz ensuring consistent high-quality data."

4. Peak vs. Onset Timing

The 0.2 ms refers to **onset detection** (start of the sound event), not peak location. The paper treats bounces as discrete events and measures how quickly the detector identifies when a bounce begins relative to actual occurrence, not when maximum energy occurs.

7. assistant (2026-06-06T17:37:04.186Z)

The source strongly corroborates the claim. Let me verify the broader context ? whether this is a controlled lab study with a curated dataset (which would qualify the generalizability) versus real match conditions, and check the dataset/setup. This matters for assessing whether the

claim's strength matches the evidence.

8. user (2026-06-06T17:37:12.680Z)

Spin Detection Using Racket Bounce Sounds in Table Tennis

Authors: Thomas Gossard, Julian Schmalzl, Andreas Ziegler, Andreas Zell (Cognitive Systems Group, University of Tuebingen, Germany)

Data Collection Method

The researchers employed a **controlled lab setup** rather than competitive match recordings. They used "the Zoom H4n Pro Handy Recorder, a directional microphone, at 44.1kHz" with "the microphone was always oriented towards the player at distances ranging from 50 cm to 2 m." The dataset was purpose-built specifically for this research rather than sourced from existing competitive audio.

Sample Size

The total dataset contained **3,396 samples** of racket-bounce sounds across 10 racket configurations, plus an additional 1,239 "other" samples (non-bounce sounds like falling objects). Training and testing used an "80/20 split ratio."

Acknowledged Limitations

The authors recognize several constraints:

- Spin direction prediction:** "predicting the direction of the spin is considerably more difficult and requires visual input"
- Quality assessment:** "One cannot judge the quality of a racket stroke from only the sound"
- Contact duration challenges:** The extremely brief contact time (1.3-1.8 ms) in table tennis makes detection "more challenging than tennis"
- Similar sounds:** Backspin and topspin sounds are "almost identical," making spin-type differentiation difficult despite detecting spin presence

9. assistant (2026-06-06T17:37:15.724Z)

I now have a thorough understanding of the source. Let me do one cross-check search to ensure there's no contradicting evidence or a known retraction/critique of this paper's temporal precision claims.

10. user (2026-06-06T17:37:24.240Z)

Web search results for query: "table tennis racket bounce sound spin detection CNN Gossard Tuebingen audio onset 0.2 ms"

Links: [{"title": "Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline This work was funded by Sony AI", "url": "https://arxiv.org/html/2409.11760"}, {"title": "Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline", "url": "https://arxiv.org/pdf/2409.11760"}, {"title": "Spin Detection Using Racket Bounce Sounds in Table Tennis", "url": "https://arxiv.org/abs/2409.11760"}, {"title": "Spin Detection Using Racket Bounce Sounds in Table Tennis", "url": "https://openreview.net/pdf?id=faYmBlDaVC"}, {"title": "(PDF) Spin Detection Using Racket Bounce Sounds in Table Tennis", "url": "https://www.researchgate.net/publication/384115010_Spin_Detection_Using_Racket_Bounce_Sounds_in_Table_Tennis"}, {"title": "Spin Detection Using Racket Bounce Sounds in Table Tennis This research was funded by Sony AI.", "url": "https://arxiv.org/html/2409.11760v1"}, {"title": "Detection of Ball Spin Direction using Hitting Sound in Tennis | Request PDF", "url": "https://www.researchgate.net/publication/346"}]

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826680_Detection_of_Ball_Spin_Direction_using_Hitting_Sound_in_Tennis"}, {"title": "(PDF) Ball Hit Detection in Table Tennis Games Based on Audio Analysis", "url": "https://www.researchgate.net/publication/224649455_Ball_Hit_Detection_in_Table_Tennis_Games_Based_on_Audio_Analysis"}, {"title": "GitHub - cogsys-tuebingen/tt_sounds: Repository for table tennis racket bounce sounds dataset · GitHub", "url": "https://github.com/cogsys-tuebingen/tt_sounds"}, {"title": "Our table tennis bounce detection and classification pipeline | Download Scientific Diagram", "url": "https://www.researchgate.net/figure/Our-table-tennis-bounce-detection-and-classification-pipeline_fig1_384115010"}]
```

Based on the search results, I found relevant research on this topic:

Summary

The research was conducted by Thomas Gossard, Julian Schmalzl, Andreas Ziegler, and Andreas Zell from the Cognitive Systems Group at University of Tuebingen.

The work introduces a real-time pipeline that tackles three core tasks: millisecond-accurate bounce detection, classification of the contact surface (racket, table, floor), and detection of spin application from audio alone.

****Technical Approach:****

Detected segments are converted to Mel spectrograms and classified using CNNs to identify the bounce surface and detect whether spin was applied. Bounce candidates are detected using energy peaks based on an exponential moving average on the high-pass filtered signal.

****Dataset:****

The team collected and annotated a high-quality dataset of 3,396 bounce recordings across 10 diverse racket configurations. The dataset and code are publicly available at https://github.com/cogsys-tuebingen/tt_sounds.

****Performance:****

The system achieves millisecond-level onset accuracy, which addresses the temporal precision needed for table tennis analysis.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

11. assistant (2026-06-06T17:37:35.066Z)

I have sufficient evidence to render a verdict. Let me assess the claim against what the source actually says.

The claim states: "A two-stage audio pipeline (energy-based peak detection followed by a small CNN) can detect racket-ball contact events from sound with near-millisecond temporal precision (energy peaks detected on average 0.2 ms after onset, even with noise), demonstrating that audio is a high-resolution local cue for hit timing in a racket sport."

Verifying each component:

- Two-stage pipeline (energy peak detection ? CNN): CONFIRMED. The paper uses energy-based peak detection (exponential moving average on high-pass filtered signal) followed by CNN classification.
- "0.2 ms after onset, even with noise": CONFIRMED. 0.09 ms no-noise, 0.2 ms with overlaid human speech.
- Racket-ball contact events: CONFIRMED. The contact (1.3-1.8 ms) is the detected event; the paper is "Sound-Based Spin Estimation in Table Tennis."
- "racket sport": CONFIRMED. Table tennis is a racket sport.
- "high-resolution local cue for hit timing": Reasonable interpretation of millisecond-level onset accuracy.

One nuance to weigh adversarially: the 0.2 ms result is from a controlled lab setup (close mic

50cm-2m, speech overlaid as the only noise ? not real crowd/match audio), and "even with noise" was tested only against overlaid human speech, not the crowd/score-call/multi-source noise of a real match. However, the claim as worded ("even with noise") is literally supported by the paper's own noisy-condition experiment. The claim does not overreach into "works in any real-world environment." The source is a primary, peer-reviewed-style arXiv paper (Sony AI funded, published dataset + code), which matches the claim's strength.

The claim is well-supported, current (Sept 2024), from a primary source, and accurately characterized. I cannot refute it.

12. user (2026-06-06T17:37:43.043Z)

Structured output provided successfully

13. assistant (2026-06-06T17:37:47.573Z)

Verdict delivered: ****refuted=false**** (claim verified, high confidence). The claim is accurately supported by the primary arXiv source (2409.11760) on every load-bearing element ? the two-stage energy-peak?CNN pipeline, the 0.2 ms onset precision under noise, and racket-ball contact detection in a racket sport (table tennis). The only caveat (controlled lab setup, noise = overlaid speech only, not real-match crowd audio) is a generalization limit but does not contradict the claim as worded.